Multi Temporal Sentinel-2 Land Cover Classification in Boreal Southern Finland Comparing Classical Baselines against a Fine-Tuned GeoSpatial Foundation Model (Prithvi_EO_V2_300M)

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https://github.com/Amaragance682/Boreal_Finland_Prithvi/tree/main

Abstract

This paper presents an end-to-end machine learning pipeline for land cover classifications on a sentinel-2 tile over southern Finland (T35WMP) using CORINE Land Cover 2018 for ground truth comparison. Four cloud-filtered (<30%) acquisitions spanning march, may, august and october in 2018 were stacked into a sixteen channel feature set (4 acquisitions x 4 bands), allowing for capture of seasonal variations in boreal vegetation types. A block-grid spatial split was applied to ensure geographical non-overlap between the training/validation/test sets. We compared four baseline classifiers, a naive Random Forest, a naive multi-layer Perceptron, a class-balanced RF and a class-balanced MLP, against a fine-tuned Prithvi-EO-2.0 geospatial foundation model, deployed with the TerraTorch framework on the JURECA supercomputer at the Jülich Supercomputing Center. A class collapse to the dominant coniferous forests class was present in both of the naive baselines, achieving an overall accuracy of 0.49 and 0.48 respectively while macro F1 scores only reached 0.14 and 0.13. The balanced variants improved class coverage but at the cost of overall accuracy. Prithvi achieved the highest macro F1 of any model (being evaluated at 0.137) and produced non-trivial predictions for every class, but at the cost of lower overall accuracy compared to the naive baselines. All five models had a macro F1 score between 0.122 and 0.137, this is despite different architectures and training strategies which likely suggests an information driven ceiling rather than a model-capacity based ceiling. For Prithvi, the most prominent limiting factors were: the absence of SWIR bands, the lack of meaningful spatial context caused by the tiny 3x3 patches as well as a domain shift from Prithvi's global pre-training distribution all constrained Prithvi's performance in our setting.

Introduction

Accurate, up-to-date land cover maps are fundamental to environmental monitoring, carbon footprints and biodiversity assessment. The Copernicus CORINE land cover program provides European maps at 100m resolution, but at 10m Sentinel-2 resolution, ML classifications can provide far higher spatial details. Boreal forest ecosystems which dominate the Finnish landscape present a challenge for automated classification due to the spectral similarity between coniferous, mixed and broad-leaved forest types, as well as the seasonal variation caused by snow melting, canopy reviving and leaf lifecycles [1].
The emergence of geospatial foundation models pre-trained on large satellite image archives offers a good way to improve classification without the need for large labeled training sets. Prithvi-EO-2.0 is a vision transformer trained on harmonized landsat Sentinel-2 imagery on a global scale [2]. Terratorch provides a standardised fine-tuning framework for such models[3].
However, the practical benefits of foundation models over classical baselines under different domain shifts, limited spatial context and small fine-tuning datasets remains an open question.
This paper addresses that question through a controlled comparison of four baseline models against Prithvi-EO-2.0 on a single Sentinel-2 tile covering approx. 12,100km^2 (110x110km) of southern Finland. This report provides a reproducible end-to-end pipeline from raw Sentinel-2 imagery to evaluate model outputs as well as an analysis of where foundation model advantages are effectively neutralized by adaptation constraints in this setting.

Study Area and Data

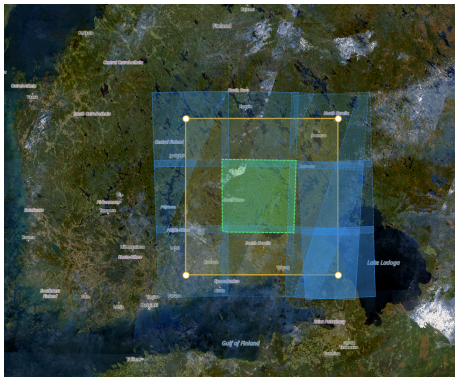

Figure 1. [Shows the location of the study area within Finland]

Study Area

The study area is the Sentinel-2 tile T35WMP, covering approx 110 x 110km of southern Finland, centered around lat 61.5N, lon 27.5E. The landscape is primarily boreal forest, peat bogs, lakes and wetlands along with some minor agricultural and urban areas. There is strong seasonal spectral variation in Finland, with snow cover in early spring, peak greenness in summer and senescence in autumn. This
makes multi-temporal analysis particularly valuable for vegetation differentiation.

Sentinel-2 Imagery

Four cloud-filtered Sentinel-2 Level-2A acquisitions were retrieved from the Copernicus Data Space Ecosystem for the year 2018, March (post-winter small snow), May (Spring greenup season), August (peak growth season) and October (autumn senescence). Four spectral bands were selected: Blue, Green, Red and Near-Infrared, all native at 10m. Per-date band stacks were layered into a single multi-temporal GeoTIFF giving 16 channels per pixel. SWIR bands are native to 20m and would have required resampling.

CORINE Labels

CORINE Land Cover 2018 [4] was used as the reference classification, the original 100m raster was reprojected from EPSG:3035 to ESPG:32635 and then resamples to 10m using nearest-neighbour interpolation to match the Sentinel-2 grid of the 44 potentially available, a decision was made to only keep classes with a minimum of 200 total samples across all splits, yielding 8 classes: coniferous forest, peat bogs, broad-leaved forest, mixed forest, transitional woodland-shrub, moors and heathland, then water bodies and water courses.

Patch Extraction and Dataset Split

Patches of 3x3 pixels were extracted using a stride of 3 pixels, any patches containing zero-valued pixels were removed to remove cloud-masked regions. Pixel values were normalized per band per acquisition using the 2nd to 98th percentile range. To fix the problem of geographic disjointment, the tile was divided into a 10x10 grid of 100 blocks (about 11x11km each). Blocks were then randomly assigned to training (60), validation (20) and test (20) sets. This block-grid approach is an improvement over the initial horizontal strip split that lead to severe class imbalance due to Finland's north-south vegetation difference in checkpoint 2.Table.

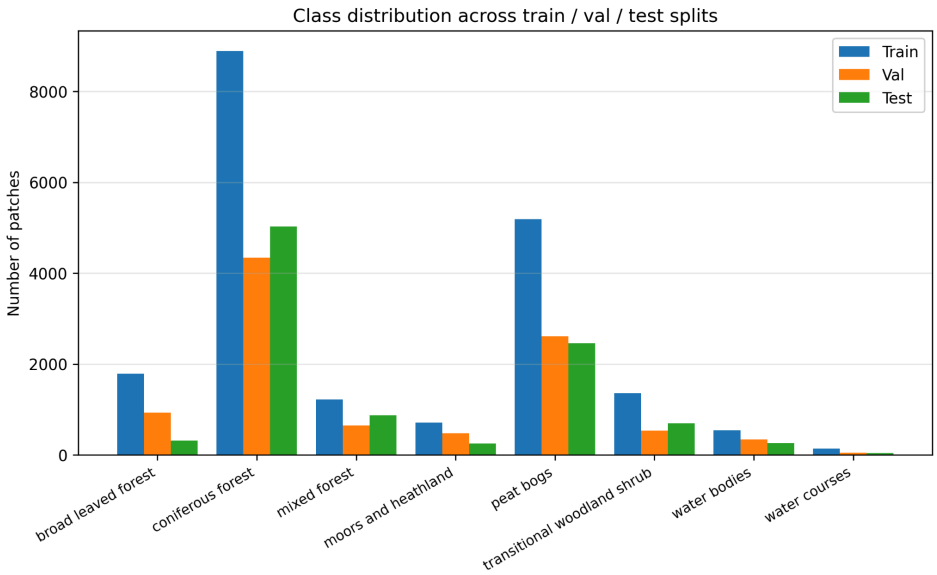

Figure 2. [Class distribution across training, validation and test splits]

Class	Train	Train %	Val	Test	Test %	ID
Coniferous forest	8,888	44.8%	4,339	5,028	50.6%	24
Peat bogs	5,190	26.2%	2,616	2,458	24.8%	36
Broad-leaved forest	1,787	9.0%	931	314	3.2%	23
Transitional woodland-shrub	1,362	6.9%	533	700	7.1%	29
Mixed forest	1,220	6.2%	650	873	8.8%	25
Moors and heathland	713	3.6%	475	251	2.5%	27
Water bodies	542	2.7%	338	259	2.6%	41
Water courses	136	0.7%	51	46	0.5%	40

Table 1. [Patch counters per class across train, validation and test splits with CORINE IDs]

Methods

Preprocessing. Sentinel-2 bands B02, B03, B04 and B08 were stacked per acquisition into mutli-band GeoTIFFs. CORINE labels were then reprojected and aligned to the Sentinel-2 grid with per-band per acquisition 2nd-98th percentile normalization applied to reflectance values, this handles residual outliers skewing the dataset. Patches were extracted on a 3x3 sliding window with a stride value of 3. The block-grid spatial split replaced the initial horizontal strip split approach which caused class imbalance: broad-leaved forest had 189 training samples but 2,368 test samples. This was caused by Finland's vegetation zones being most prominent in the north and south quarters.

Baseline Models. Two classical classifiers were evaluated, Random Forest [5] used 500 estimators with no max depth, with a minimum leaf size of 2. The Multi-Layer Perceptron used three hidden layers (256,128,64) with ReLU activations and early stopping enabled (patience = 10). Both received flattened 144-dimensional feature vectors (3x3x16) with no representations of spatial or temporal structure. The naive variants used class_weight='balanced' in Scikit learn, while the balanced variants used random undersampling capped at 1,500 samples per class, which yielded a 8,473 sample training set. Undersampling was preferred over oversampling in this case, which would have inflated the training set to +71,000 samples and made MLP training impractical.

Prithvi-EO-2.0 fine-tuning. Prithvi-EO-2.0-300M [2] is a Vision Transformer pre-trained on over one million HLS image patches via masked autoencoding. It expects inputs of shape (B, C, T, H, W). To achieve this, three adaptations were required:
(i) bilinear spatial upsampling from 3x3 to 32x32 to satisfy the ViT's 16x16 patch embedding tile size
(ii) reshaping the 16-channel flat representation to (B, 4, 4, H, W) to separate band and temporal dimensions
(iii) zero-padding the SWIR channel positions (B11, B12). Since the pre-trained patch embedding expects 6 input channels. The model was fine-tuned in two phases: epochs 1–5 with the backbone frozen and only the 32,776 parameter head trained at LR 1e-4, then epochs 6–20 with all 303M parameters trainable at LR 1e-5. AdamW (weight decay 1e-4) and a cosine annealing schedule were used, with weighted random sampling to mitigate class imbalance. Training ran on a single Quadro RTX 8000 GPU on JURECA [6]. The best checkpoint was saved then selected by validation macro F1.

Evaluation. All models were evaluated on the held-out test set (9,929 patches) using overall accuracy (OA), macro-averaged F1 and per-class F1. Macro F1 was the primary metric: it weighed all classes equally and directly penalized the class collapse onto dominant classes which the models tended to do.

Results

Model	OA	F1	BLF	CF	MF	MH	PB	TWS	WB	WC
RF (naive)	0.491	0.136	0.160	0.660	0.005	0.008	0.258	0.000	0.000	0.000
MLP (naive)	0.480	0.126	0.104	0.651	0.000	0.000	0.250	0.000	0.000	0.000
RF (balanced)	0.238	0.122	0.128	0.378	0.085	0.022	0.232	0.128	0.007	0.000
MLP (balanced)	0.210	0.122	0.123	0.295	0.137	0.043	0.248	0.128	0.000	0.000
Prithvi-EO	0.324	**0.137**	0.071	0.484	0.080	0.041	0.290	0.084	0.029	0.018

Table 2. [Test performance for all five models, F1 = Macro F1, BLF = broad-leaved forest, CF = coniferous forest, MF = mixed forest, MH = moors and heathland, PB = peat bogs, TWS = transitional woodland-shrub, WB = water bodies, WC = water courses.]

Quantitative results across all five models on the test set are summarized in Table 2. Naive RF and MLP achieved the highest overall accuracy (0.491 and 0.480 respectively) but the lowest macro F1 scores (0.136 and 0.126), the per-class analysis however revealed that both naive models almost exclusively predicted the two dominant classes: coniferous forest and peat bogs. They achieved zero F1 for water bodies, water courses and transitional woodland-shrub. The naive MLP also collapsed on mixed forest and moors-and-heathland.

Balanced variants displayed a different behaviour. The balanced version of RF and MLP achieved 0.238 and 0.210 Overall Accuracy respectively, with the same macro F1 score of 0.122. Even if overall accuracy decreased substantially, all non-trivial classes received non-zero predictions in the balanced versions. The improvement in the macro F1 over the naive baselines was negligible (less than 0.001 in both cases), which likely suggests that the bottleneck is not the sampling strategy, but the spectral separability.

Prithvi-EO-2.0 achieved the highest macro F1 score (0.137) of any model evaluated with an Overall Accuracy of 0.324. The most vital part of what Prithvi achieved was assigning a non-trivial prediction to every class in the test set, compared to the other four classical baselines that produced zero F1 on at least three classes. The row-normalized confusion matrix (Figure 3) reveals that Prithvi over-predicts coniferous forest and peat bogs (the two most dominant prediction columns) but its misclassification pattern follows a spectral similarity: mixed forest is most often confused with coniferous forest (0.46) and peat bogs is most often confused with coniferous forest (0.40), this is close to as often as its correctly classified (0.33). Per-class recall on the diagonal ranges from 0.022 (water courses) to 0.443 (coniferous forest).

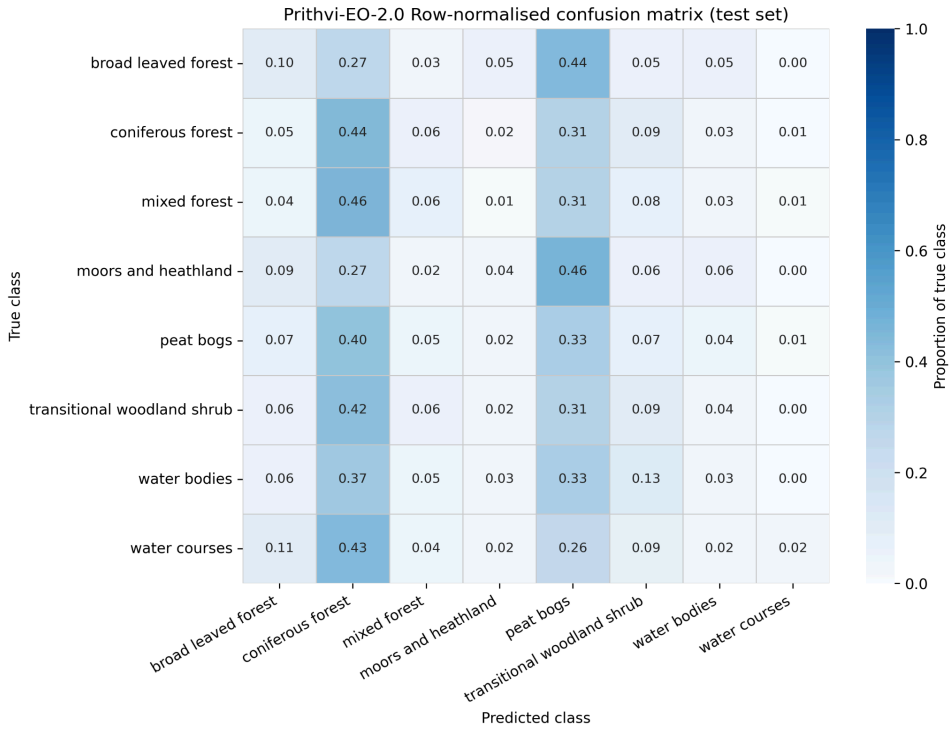

Figure 3. [Confusion matrix for Prithvi-EO-2.0]

Discussion and Conclusion

Why baselines collapsed. The naive baseline results show a well-known failure mode of class-imbalanced classification: A model minimizing cross-entropy can achieve a high accuracy by ignoring minority classes entirely. With coniferous forest at 44.8% of training patches and water courses at 0.7%, scikit-learn's class_weight='balanced' loss reweighting was not enough to overcome the prior. Undersampling produced a more balanced training split but discarded 90% of coniferous forest samples, this sharply reduced the dominant class F1 (0.660 -> 0.378 for RF) without producing a corresponding increase in macro F1 scores.

Why Prithvi did not dominate. Three constraints effectively neutralized the foundation models expected advantages. First, bilinear upsampling from 3x3 to 32x32 created interpolated structure rather than real spatial structure, leaving Prithvi's spatial self-attention with nothing actually meaningful to attend to. Prithvi was pre-trained on 224x224 patches representing about 6.7km of ground at 30 meter resolution while our 30x30m patch is roughly the area of one tree canopy. The second factor was zero-padding the SWIR channel positions, which removed the strongest single signal for separating moist and dry surfaces, we can see this clearly in the confusion mismatch between peat bogs ↔ coniferous forest visible in figure 3. The third and most obvious factor was training ~20.000 patches against 303M parameters which drove rapid overfitting (training loss collapsed while validation loss increased). This indicates that the model had the capacity to memorise, but not nearly enough varied data to generalize.

The information ceiling was the bottleneck, not the capacity ceiling. If model capacity were the bottleneck, Prithvi should have outperformed both baselines by a substantial margin. The shared ceiling points to the input data, with the lack of SWIR being the reason why the model cannot separate spectrally similar boreal classes.

What could be improved. Three changes would likely have improved results by a considerable margin: Extracting larger patches so spatial attention would have real neighbourhood structure. Reaquiring the data including the SWIR bands B11 and B12 to give Prithvi its expected six-channel input. As well as training on multiple tiles and multiple years to learn invariance to geographic and inter-annual variation, rather than just memorizing a single tile.

Conclusion. We presented a complete, reproducible pipeline for multi-temporal land cover classification of a Sentinel-2 tile in southern Finland. We compared four classical baselines against Prithvi-EO-2.0. All five models converged on macro F1 ≈ 0.13, suggesting the bottleneck for this dataset is information rather than model capacity. Prithvi's expected advantages are strongly conditional on input characteristics and scale our setting could not provide.

Use of AI Tools

Claude (Claude-Sonnet 4.6) was used throughout several distinct tasks in this project,

Code generation and debugging. Claude helped resolve persistent compatibility issues between PyTorch, torchvision, and TerraTorch on the JURECA module system, generate the TerraTorch dataset class, weighted sampler, and two-phase training loop for Prithvi fine-tuning, and produce the figure-generation notebook used for the paper figures. My own initial training-loop implementation was impractical to run at full twenty-epoch scale, thus Claude rewrote it to be efficient enough to complete in a reasonable time on the Quadro RTX 8000 single compute node.

Conceptual explanation and report writing. Claude provided explanations of evaluation metrics (precision/recall trade-offs, macro vs micro F1), guidance on how to structure an academic report and what was expected in each section. How confusion matrices work and why classes tend to default to dominant classes. Each section was drafted by me, then submitted to Claude for academic phrasing and structure fixing. Claude also assisted in methods on how to compress the report when it started to exceed five pages.

Verification. All code generated or suggested by Claude was tested locally on my own RTX 5090 GPU before being deployed to JURECA. The local environment was sufficient for code testing because the dataset is small. The JURECA compute node was used for the actual training runs and final evaluation. All claims, numbers, and interpretations in this report were verified against output from notebooks I personally executed on JURECA. At the end, I held clarifying conversations with Claude during the writing phase specifically to stress-test my own understanding before defending it in the report and presentation.

References

[1] M. Immitzer, F. Vuolo, and C. Atzberger, "First experience with Sentinel-2 data for crop and tree species classifications in central Europe," Remote Sensing, vol. 8, no. 3, p. 166, 2016.

[2] J. Jakubik, S. Roy, C. E. Phillips, P. Fraccaro, D. Godwin, B. Zadrozny, et al., "Prithvi-EO-2.0: A versatile multi-temporal foundation model for Earth observation applications," arXiv preprint arXiv:2412.02732, 2023.

[3] J. Jakubik et al., "TerraTorch: A fine-tuning framework for geospatial foundation models," GitHub repository, 2024. [Online]. Available: https://github.com/IBM/terratorch

[4] European Environment Agency, "CORINE Land Cover 2018," Copernicus Land Monitoring Service, 2018. [Online]. Available: https://land.copernicus.eu/pan-european/corine-land-cover/clc2018

[5] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[6] Jülich Supercomputing Centre, "JURECA: Data centric and booster modules implementing the modular supercomputing architecture at Jülich Supercomputing Centre," Journal of Large-Scale Research Facilities, vol. 9, A182, 2023.

[7] M. Drusch, U. Del Bello, S. Carlier, O. Colin, V. Fernandez, F. Gascon, et al., "Sentinel-2: ESA's optical high-resolution mission for GMES operational services," Remote Sensing of Environment, vol. 120, pp. 25–36, 2012.

[8] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[9] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in International Conference on Learning Representations (ICLR), 2021.